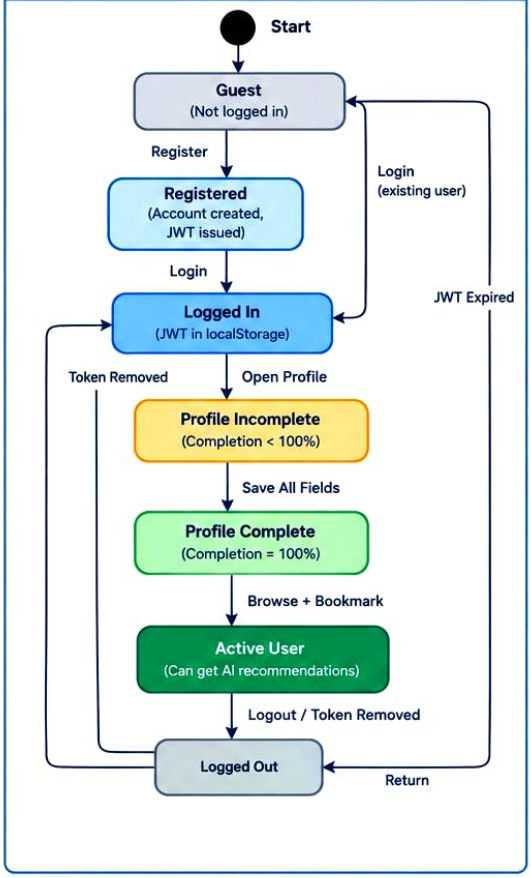
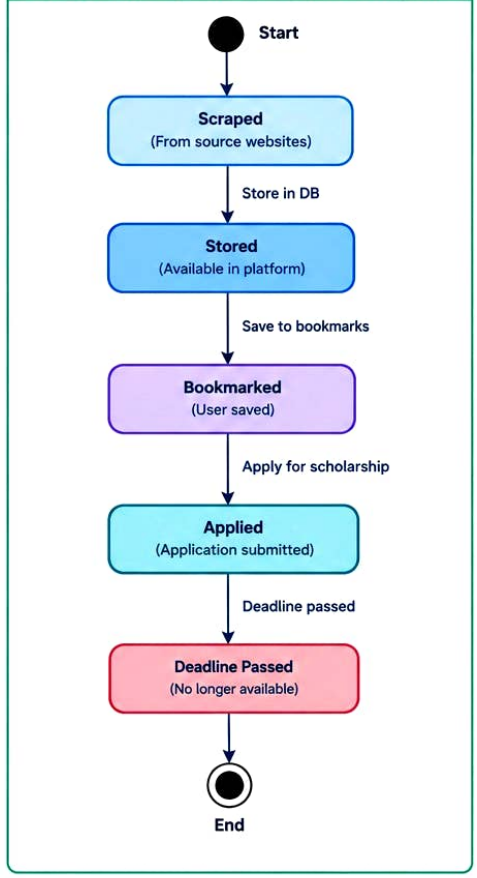
ScholarNexus — State Diagrams

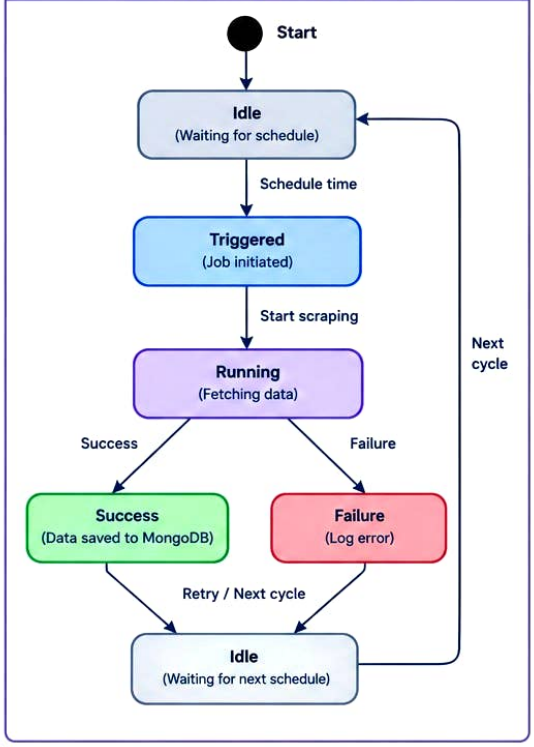
1. User Session State Diagram



2. Scholarship Lifecycle State Diagram



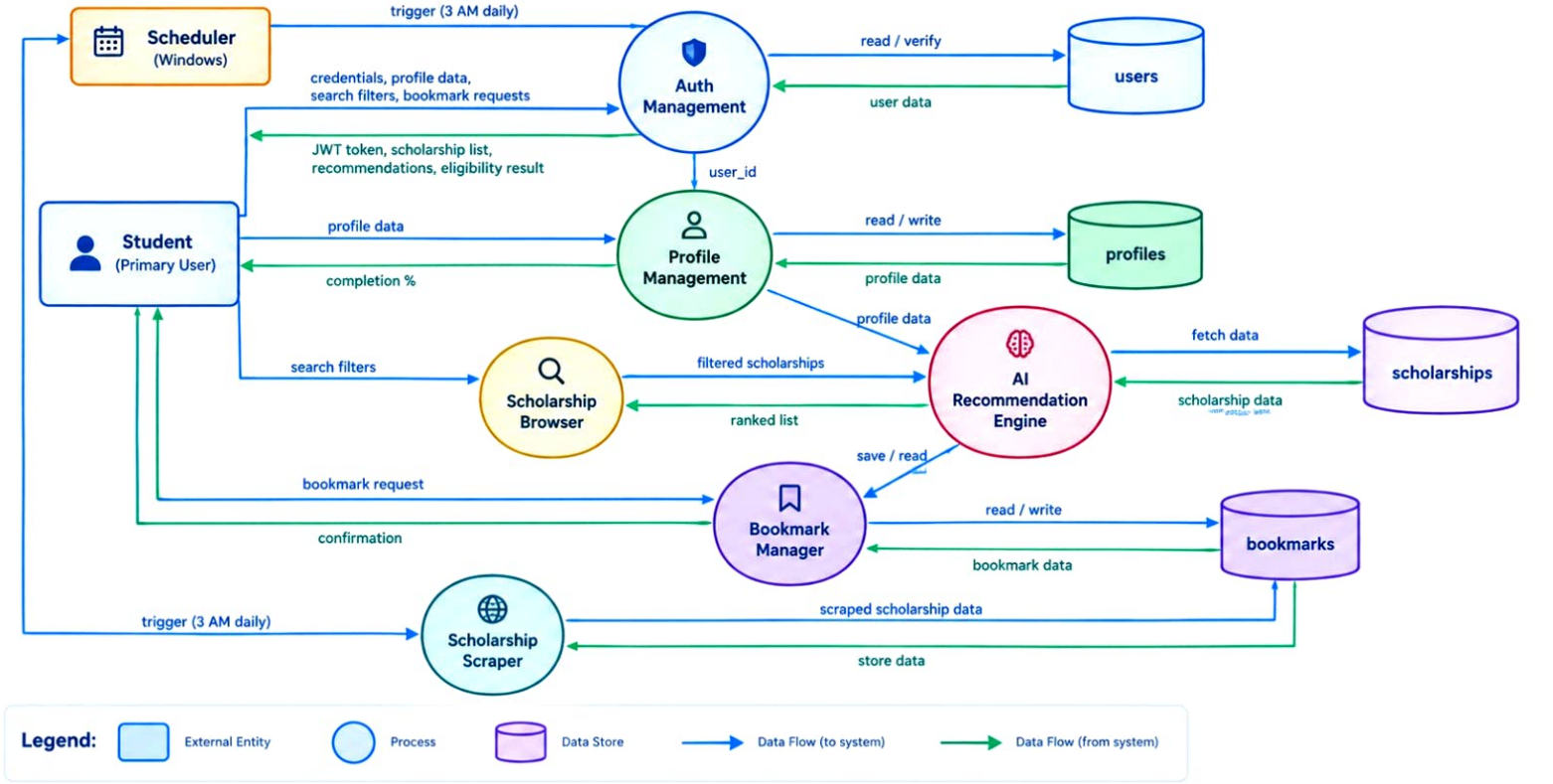
3. Scraper Job State Diagram

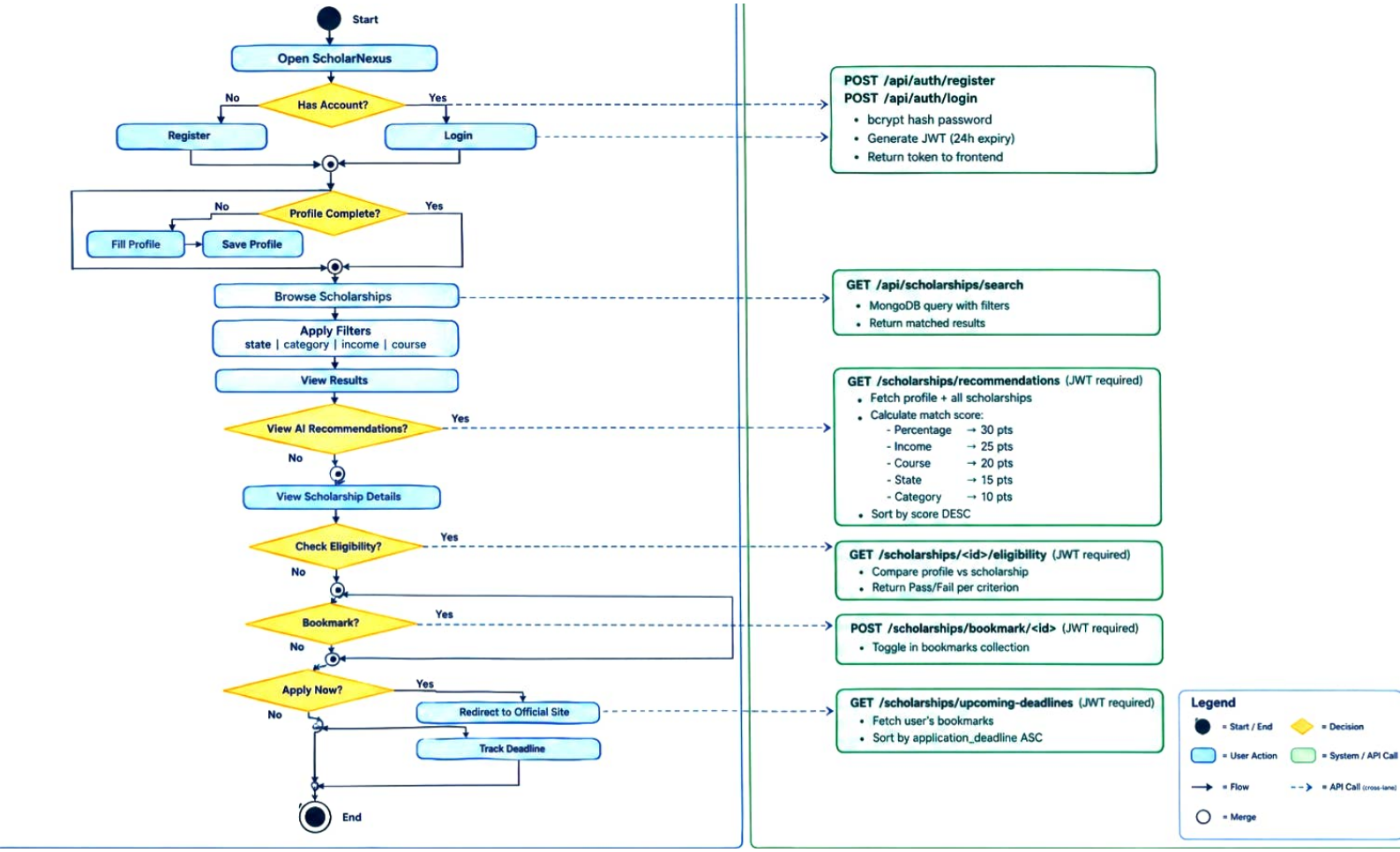


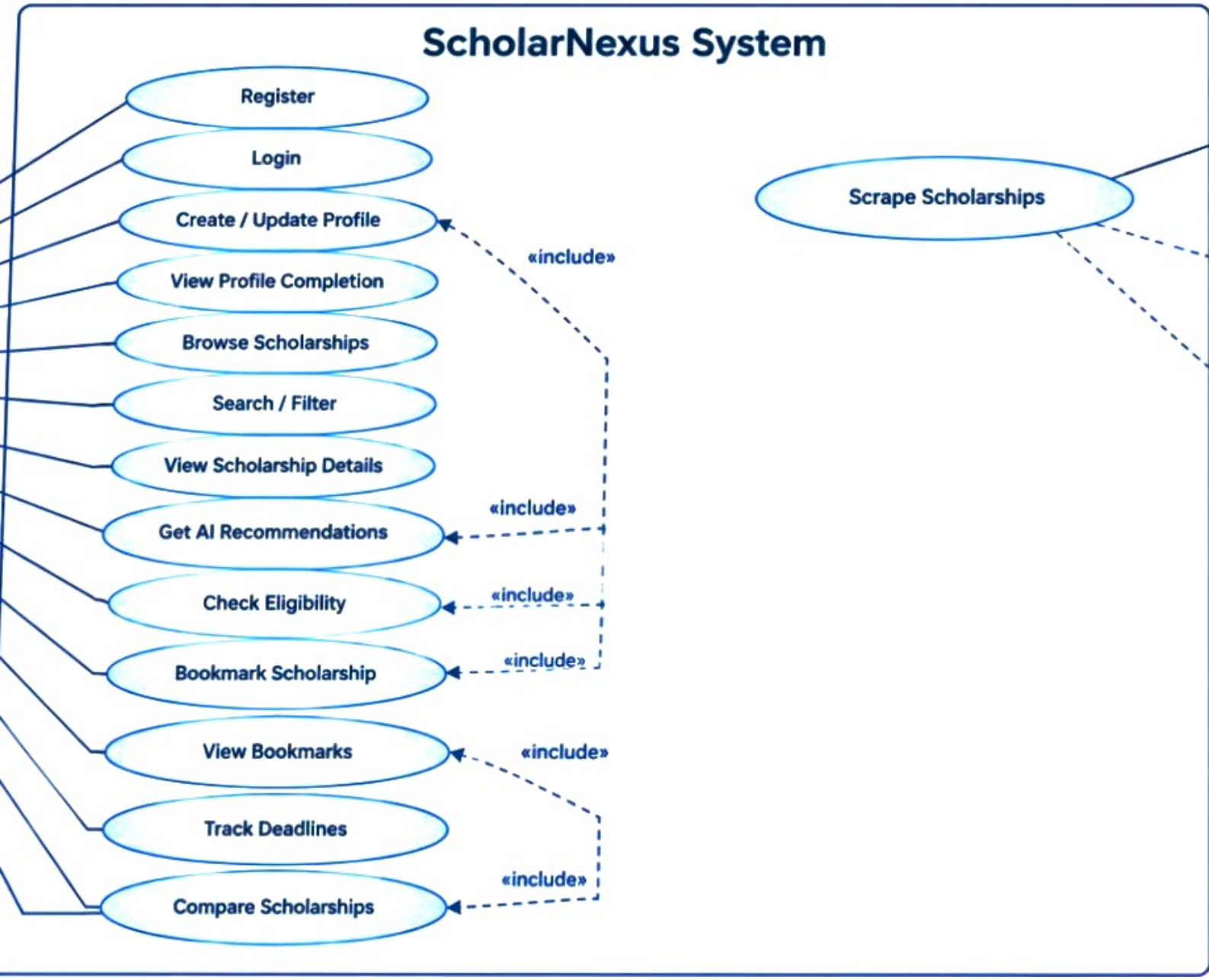
Legend:



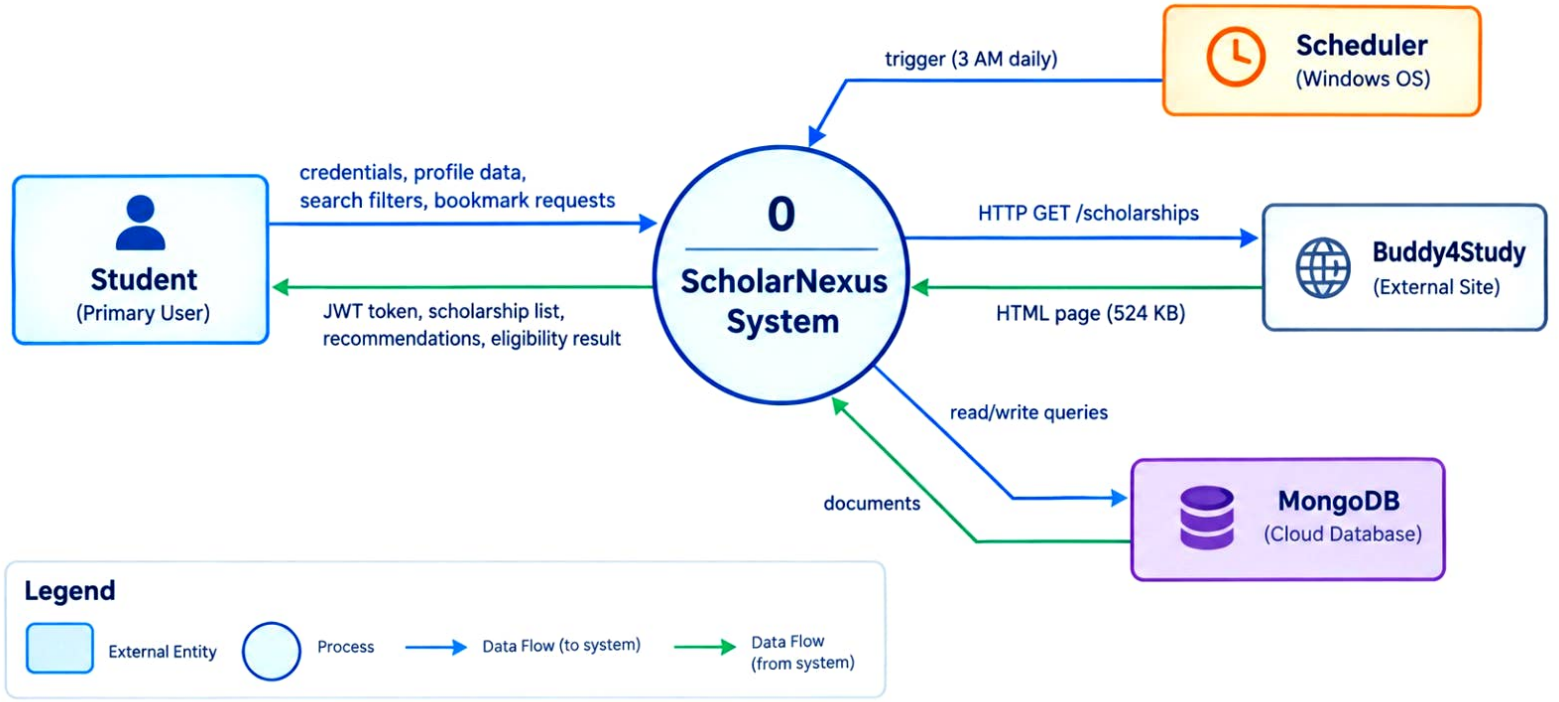
ScholarNexus — DFD Level 1

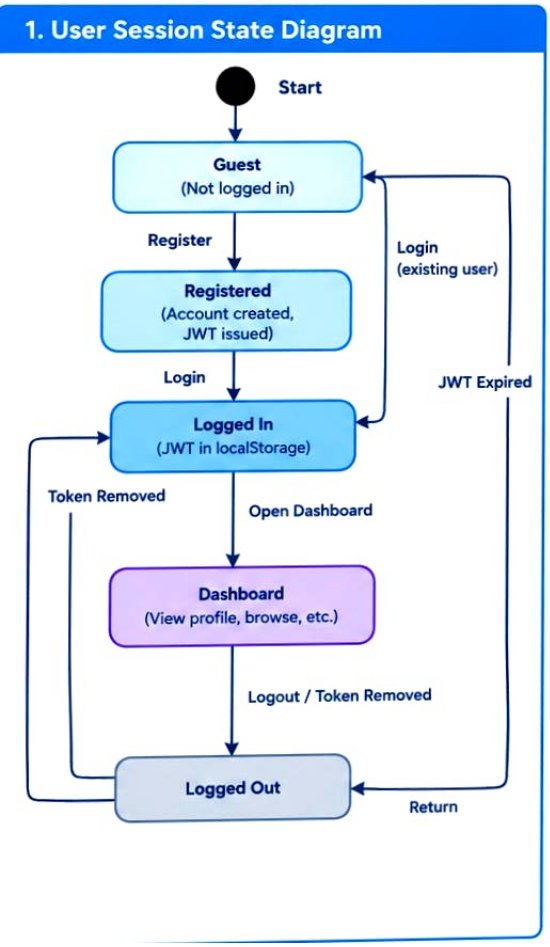


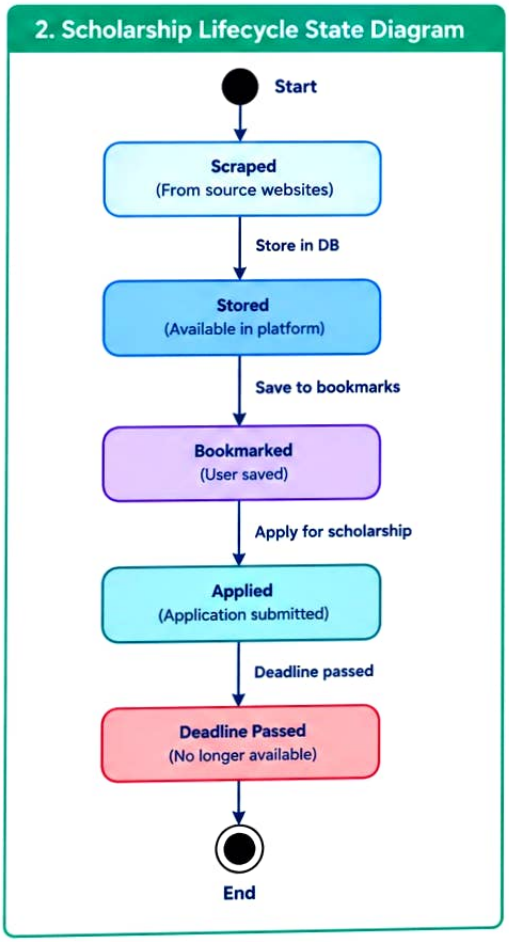

Start
Open ScholarNexus
Has Account?
No
Yes
Register
Login
Profile Complete?
No
Yes
Fill Profile
Save Profile
Browse Scholarships
Apply Filters
state | category | income | course
View Results
View AI Recommendations?
Yes
No
View Scholarship Details
Check Eligibility?
Yes
No
Bookmark?
Yes
No
Apply Now?
Yes
No
Redirect to Official Site
Track Deadline
End
POST /api/auth/register
POST /api/auth/login
• bcrypt hash password
• Generate JWT (24h expiry)
• Return token to frontend
GET /api/scholarships/search
• MongoDB query with filters
• Return matched results
GET /scholarships/recommendations (JWT required)
• Fetch profile + all scholarships
• Calculate match score:
- Percentage → 30 pts
- Income → 25 pts
- Course → 20 pts
- State → 15 pts
- Category → 10 pts
• Sort by score DESC
GET /scholarships/<id>/eligibility (JWT required)
• Compare profile vs scholarship
• Return Pass/Fail per criterion
POST /scholarships/bookmark/<id> (JWT required)
• Toggle in bookmarks collection
GET /scholarships/upcoming-deadlines (JWT required)
• Fetch user's bookmarks
• Sort by application_deadline ASC
Legend
= Start / End
= Decision
= User Action
= System / API Call
= Flow
= API Call (cross-lane)
= Merge

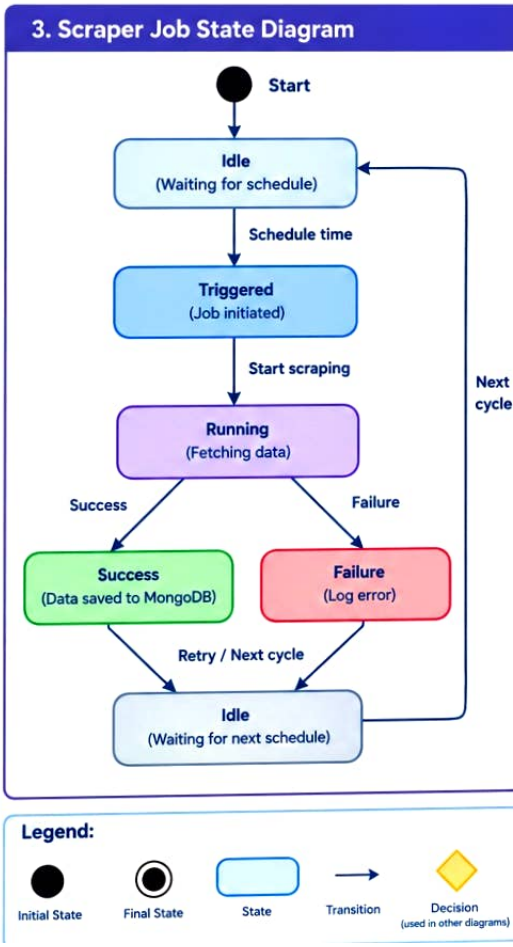

ScholarNexus System
Register
Login
Create / Update Profile
View Profile Completion
Browse Scholarships
Search / Filter
View Scholarship Details
Get AI Recommendations
Check Eligibility
Bookmark Scholarship
View Bookmarks
Track Deadlines
Compare Scholarships
Scrape Scholarships
«include»
«include»
«include»
«include»
«include»
«include»

ScholarNexus — DFD Level 0 (Context Diagram)




1. User Session State Diagram
Start
Guest
(Not logged in)
Register
Registered
(Account created,
JWT issued)
Login
Logged In
(JWT in localStorage)
Token Removed
Open Dashboard
Dashboard
(View profile, browse, etc.)
Logout / Token Removed
Logged Out
Return
Login
(existing user)
JWT Expired


2. Scholarship Lifecycle State Diagram
Start
Scraped
(From source websites)
Store in DB
Stored
(Available in platform)
Save to bookmarks
Bookmarked
(User saved)
Apply for scholarship
Applied
(Application submitted)
Deadline passed
Deadline Passed
(No longer available)
End


3. Scraper Job State Diagram
Start
Idle
(Waiting for schedule)
Schedule time
Triggered
(Job initiated)
Start scraping
Running
(Fetching data)
Success
Failure
Success
(Data saved to MongoDB)
Failure
(Log error)
Retry / Next cycle
Idle
(Waiting for next schedule)
Next cycle

Legend:
Initial State
Final State
State
Transition
Decision
(used in other diagrams)